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Evolution of olfactory sensitivity, preferences and behavioral responses in Mexican cavefish: fish personality matters

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Abstract

Animals are adapted to their natural habitats and lifestyles. Their brains perceive the external world *via* their sensory systems, compute information and generate appropriate behavioral outputs. However, how do these processes evolve across evolution? Here, focusing on the sense of olfaction, we have studied the evolution in olfactory sensitivity, preferences and behavioral responses to six different amino acid odors in the two morphs of the fish *Astyanax mexicanus*. To this end, we have developed a high-throughput behavioral setup and pipeline of quantitative and qualitative behavior analysis, and we have tested 425 six-week-old *Astyanax* larvae. The blind, dark-adapted morphs of the species showed markedly distinct basal swimming patterns and behavioral responses to odors, higher olfactory sensitivity and a strong preference for alanine, as compared to their river-dwelling eyed conspecifics. In addition, we discovered that fish have an individual "swimming personality", and that this personality influences their capability to respond efficiently to odors and find the source. Importantly, the personality traits that favored significant responses to odors were different in surface fish and cavefish. Moreover, the responses displayed by second-generation cave x surface F2 hybrids suggested that olfactory-driven behavior is a quantitative genetic trait. Our findings show that olfactory processing has rapidly evolved in cavefish at several levels: detection threshold, odor preference, and foraging behavior strategy. Cavefish is therefore an outstanding model to understand the genetic, molecular and neurophysiological basis of sensory specialization in response to environmental change.

eLife assessment

In this **important** paper, Blin and colleagues develop a high-throughput behavioral assay to test spontaneous swimming and olfactory preference in individual Mexican cavefish larvae. The authors present **compelling** evidence that the surface and cave morphs of the fish show different olfactory preferences and odor sensitivities and that individual fish show substantial variability in their spontaneous activity that is relevant for olfactory behaviour. The paper will be of interest to neurobiologists working on the evolution of behaviour, olfaction, and the individuality of behaviour.

Introduction

With more than 26.000 species representing half of vertebrates, bony fishes are extremely diverse and have colonized all possible ecological niches (Helfman et al., 2009), making them outstanding models to study the neurophysiological, genetic and evolutionary underpinnings of adaptive behaviors. To colonize, survive and thrive in various environments, sensory systems are particularly crucial as they serve as windows to the external world, and they display an exceptional diversity in fishes. From the visual system of deep-sea fish shaped to catch the rare photons or the point-like bioluminescent signals (de Busserolles et al., 2020) to the mobile barbels covered with taste buds to probe the sea bottom and detect buried prey of bottom feeders (Kiyohara et al., 2002), examples abound.

The anatomical sensory specializations, that is the relative importance taken by specific sensory organs or brain sensory areas, arise during embryonic development through regulatory processes that control the patterning of the neuroepithelium and that define boundaries between presumptive neural regions (Krubitzer et al., 2011). The relative investment in different sensory modules governs anatomical and behavioral specializations. Among fish, sand-dwelling cichlids that rely almost exclusively on vision to execute behaviors develop a large optic tectum, whereas rock-dwelling cichlids, which inhabit complex environments, have small optic tecta but an enlarged telencephalon, and these variations arise early in development (Sylvester et al., 2010). Similarly, in the early embryos of dark-adapted blind Mexican cavefish, the presumptive eyefield territory is reduced but the presumptive olfactory epithelium is increased in size, presumably as a compensation for the loss of visual modality in the dark environment (Agnès et al., 2022; Hinaux et al., 2016a; Pottin et al., 2011; Torres-Paz et al., 2019).

While the comparative anatomy of fish brains is reasonably well documented, behavioral studies have mainly focused on a few model species, primarily zebrafish (Kalueff et al., 2013). The behavioral correlates of sensory systems diversity and evolution in fishes are poorly described and understood, which hampers the interpretation of cross-species comparisons. For example, the size and complexity of fish olfactory organs is highly variable and correlates with the richness of olfactory receptors repertoire, defining a morpho-genomic space in which olfactory specialists and non-specialists species distribute (Burguera et al., 2023; Policarpo et al., 2022). Yet, the differences in olfaction-driven behaviors, olfactory sensitivity and olfactory preferences between sunfish or pipefish, which have a small and flat olfactory epithelium and possess ~30 olfactory receptor genes and polypteriforms, which have large and complex olfactory rosettes with up to 300 lamellae and possess ~1300 olfactory receptor genes, are completely unknown.

To start addressing the question of the evolution of olfactory sensory-driven behaviors in an amenable laboratory fish model, we used the blind cave and the river dwelling morphs of the Mexican tetra, the characiform *Astyanax mexicanus*. The species has become an established model for evolutionary biology at large, including evolution of behaviors (Duboué and Keene, 2016; Hinaux et al., 2016b; Kowalko, 2020; Yoshizawa, 2016). Cavefish embryos and larvae have larger olfactory pits than surface fish and their neuronal composition is changed (Blin et al., 2018; Hinaux et al., 2016a), although adults have a similar olfactory rosette with 20-25 lamellae (Schemmel, 1967). The recent divergence between the two forms probably did not allow for a substantial evolution of their olfactory receptor gene repertoire, which is almost identical in the two eco-morphotypes (245 genes in cavefish, 233 genes in surface fish, from recent genome assemblies) (Policarpo et al., 2022). Yet, cavefish larvae tested in groups display outstanding olfactory detection capacities, as they are attracted to extremely low concentration of the amino acid alanine (10^{-10}M) while surface fish larvae have a more "classical" threshold (10^{-5}M) (Hinaux et al., 2016a) which approximates levels of free amino acids found in natural waters (Hara, 1994). In wild natural caves too, groups of cavefish adults respond to low concentrations of odorants (Blin et al., 2020). We hypothesized that the evolution of olfactory system development and olfactory skills in blind cavefish is an adaptive trait that, together with other constructive sensory changes in mecano-sensory and gustatory systems (Varatharasan et al., 2009; Yamamoto et al., 2009; Yoshizawa and Jeffery, 2011; Yoshizawa et al., 2014), may have contributed to their adaptation and survival in the extreme environment of their subterranean habitat. Focusing on olfaction, here we developed a sensitive, high-throughput behavioral assay and a pipeline for analysis to compare the types of behavioral responses elicited by food-related odors in cavefish and surface fish, at population and individual levels. We determined their olfactory preferences and sensitivity thresholds for six amino acids, and we analyzed their behavioral responses in detail. We also described olfactory-driven behaviors in second-generation (F2) hybrids resulting from crosses between surface and cave morphs, as an attempt to understand the genetic architecture of the "olfaction trait" in *Astyanax* morphs. We discovered that the behaviors triggered by odorant stimulation has markedly evolved in cavefish.

Results

A high-throughput, sensitive behavioral assay to compare individual's behaviors in *Astyanax* morphotypes

Our goal was to compare 1) olfactory discrimination capacities, 2) odor preferences and 3) behavioral responses to olfactory stimuli in Pachón cavefish (CF), surface fish (SF) and F2 hybrids (F2) individuals. As the three types of fish are markedly different in terms of basal swimming activity and patterns, we first sought to characterize the diversity of these behaviors in order to be able to interpret accurately fish responses to odors.

Six weeks old fish were habituated either 1h or 24h in their individual test box and they were first recorded for 30 minutes in the dark, without any stimulus (**Figure 1**; see Methods). We observed and categorized several typical and distinctive "baseline" swimming behaviors for individual fish (**Figure 2A**): random/haphazard swim (R), wall following (WF), large or small circles (C), thigmotactism (T, along the X- or the Y-axis of the box), or combinations thereof. The distribution of these different types of swimming patterns was significantly different in Pachón CF, SF and F2 types of fish (**Figure 2B**; Fisher's exact test). A majority of SF swam in a random pattern (blue shades), while a majority of CF performed wall following (red/brown shades) and F2 fish showed more diversified swimming patterns. Importantly, the distribution of these baseline-swimming patterns was affected as a function of the habituation time, in all three types of fish (**Figure 2B**; p=0.044 for SF, p=0.0005 for CF and p=0.0005 for F2, Fisher's exact tests). Moreover, thigmotactism behavior, which we have previously shown to represent stress behavior (Pierre et al., 2020), was frequent in SF after 1h habituation (70% of individuals) but reduced after 24h acclimation (37%). In the same line, the swimming speed was different for all fish types when computed after 1h or after 24h of habituation (**Figure 2C**). This suggested that natural, unperturbed behavior is better observed after long, 24h habituation, which we applied thereafter.

As described above at population level, cavefish, surface fish and F2 displayed overall markedly different basal swimming patterns (**Figure 2B**). Yet, at individual level, there was a substantial degree of variability across individuals within a morphotype. That is, even though most SF tended to swim randomly (75% after 24h habituation; blue shades on graphs) and most CF performed wall following (50%; red/brown shades), these behaviors were not exclusive as some SF individuals displayed thigmotactism behavior or some CF individuals swam in circles. Moreover and importantly, the type of swim pattern of each individual fish was remarkably stable and reproducible along several days of recordings (**Figure 2D** and Suppl. Fig1A). Together, these observations strengthened the importance of 1) using long habituation times, and 2) studying

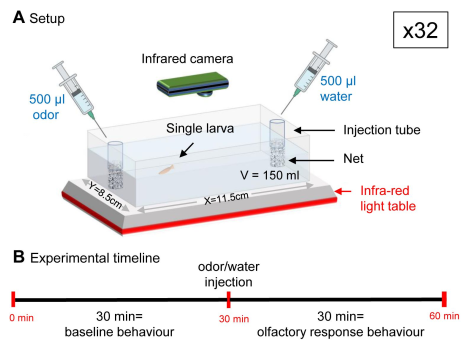


Figure 1

experimental setup for testing behavioral responses to olfactory stimulation in individual, 6 weeks old, *Astyanax mexicanus* larvae.

A, Each fish is placed individually in 150ml water in a rectangular testbox placed on an infrared light emitting table. Control or odorant solution are delivered at the two extremities of the box, inside tubes covered with a net. Medium to high throughput behavioral testing is achieved by parallel recording of 32 testboxes placed under 8 infrared recording cameras (4 testboxes per camera).

B, After one hour or 24 hours habituation, the test consists of one hour IR video recording. The first 30 minutes provide the control/baseline behavior of individual fish. Odorant stimulation is given at 30 minutes, and the behavioral responses are recorded for 30 more minutes.

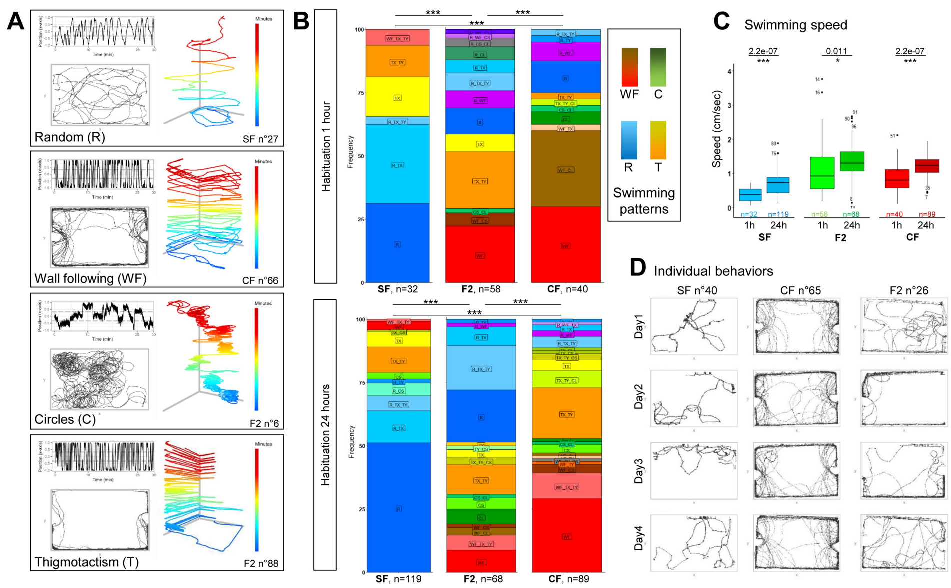


Figure 2

basal swimming behaviors of 6 week-old CF, SF and F2.

A, Examples of typical and distinctive basal swimming patterns exhibited by *Astyanax mexicanus* morphotypes. The behavioral patterns (Random, Wall Following, Circles, Thigmotactism) and the individual fish shown are indicated. Each pattern is best described by the combination of 3 types of representations. (1) the top left track shows the displacement of the fish along the x-axis of the box, representing back-and-forth swims along its long dimension. (2) the bottom left track is the top view, 2D representation of the trajectory. (3) the right track is the 3D plus time (color-coded) representation and helps, for example, to discern between wall following and thigmotatism.

B, Distribution of basal swimming pattern displayed by SF, F2, and CF after 1h or after 24h of habituation. Numbers of fish tested are indicated. The general color code is indicated, exact swimming patterns and combinations are given on the colored plots. Fisher's exact tests for statistical comparisons between groups.

C, Boxplots showing the swimming speed in SF (blue), F2 (green) and CF (red), after 1h or 24h of habituation. Values are mean speed calculated over a period of 15 minutes. Numbers of fish tested are indicated. Mann-Whitney tests p-values are shown.

D, Examples of the stability of the basal swimming pattern over 4 experimental days in 3 individuals, with 24h of habituation time. One SF displaying Random pattern, one CF displaying Wall Following and one F2 displaying Thigmotactism + Random swim are shown.

individual behaviors. They further highlighted an overlooked aspect of fish behavioral analyses: fish may have a "personality" that we sought to take into account when comparing behavioral responses to odors below.

Finally, Pachón CF, SF and F2 hybrids possess markedly different sensory apparatuses and capacities, not only for chemo-but also for mechano-sensation (Lunsford et al., 2022; Yoshizawa et al., 2010). Therefore, to test behavioral responses specific to olfactory stimuli, we wanted to find a way to deliver odors that would be vibration-less. We choose to inject odorant solutions inside tubes attached at the two extremities of the test box, designed to stop the wave and flow that would otherwise arise at the surface of the water upon injection (**Figure 1**). The efficacy of the device was demonstrated by the lack of perturbation in swimming behavior, neither qualitative (i.,e., behavioral response and swimming pattern; **Figure 3AB**) nor quantitative (i.,e., swimming kinetics; **Figure 3C**) for the 3 types of fish when water/control injections were performed on one side of the test box (or two sides; Suppl. Fig1BC). This suggested that our way to deliver odorant solutions did not generate vibratory perturbations. Moreover, when no injection was made, individual fish behaviors were stable over one hour of test (Suppl. Fig1D), allowing to observe accurately the effects of an olfactory stimulus if applied after 30 minutes. Finally, all recordings were performed in the dark under infra-red lights to neutralize the visual modality in sighted fish, a procedure which did not affect SF behavior (Suppl. Fig1E). Hence, the behavioral responses to odors recorded thereafter in Pachón CF, SF and F2 are driven by olfaction and can be compared (see also (Hinaux et al., 2016a)).

Behavioral responses to alanine in *Astyanax* morphotypes

Amino acids are food-related odorant cues for fish. Next, we characterized individual behavioral responses elicited by different concentrations of alanine, a potent aliphatic amino acid cue for most fish species including *Astyanax*.

Injection of alanine 10^{-2}M, 10^{-3}M or 10^{-4}M (hence 10^{-4}M, 10^{-5}M or 10^{-6}M final concentration in the odorant third of the testing box, respectively) induced a strong behavioral response in cavefish (**Figure 4A**, 10^{-4}M injection shown; Suppl. Fig2A for other examples). Upon stimulus, CF decreased the numbers of back-and-forth swims along the X- and the Y-axes of the box (**Figure 4A**, top left graph and **Figure 4FG**), and they shifted and restricted their swimming activity towards the odorant side of the arena, close to the odor delivery tube (**Figure 4A**, bottom left graph, and **Figure 4E**). A tendency to decreasing swimming speed was observed but was significant only for 10^{-3}M injections (**Figure 4D**). Alanine injection also markedly changed CF swimming patterns, as they completely switched from WF and T to R and C swimming modes for the three concentrations of alanine (**Figure 4H** and Suppl. Fig2D; Fisher's exact tests p values: 0.014, 0.0005 and 0.0005 for 10^{-2}M, 10^{-3}M or 10^{-4}M alanine injections, respectively). In sum, CF individuals displayed robust behavioral responses and attraction to alanine, including at the lowest concentration tested.

Conversely, surface fish responses were more subtle and seemed restricted to some individuals (**Figure 4B**; Suppl. Fig2B for other examples). Notably, the numbers of back-and-forth swims along the X- and the Y-axes of the box were mostly unchanged and the position in the box did not vary for SF upon alanine injection (**Figure 4B**, top and bottom left graphs and **Figure 4EFG**). The swimming speed was unchanged (**Figure 4D**). Swimming patterns were globally unaffected upon alanine injection (**Figure 4H** and Suppl. Fig2D; Fisher's exact tests p values: 0.16, 0.54 and 0.80 for 10^{-2}M, 10^{-3}M or 10^{-4}M alanine injections, respectively). As shown on Suppl. Fig2B, some rare SF individuals appeared to change their swimming mode upon alanine 10^{-2}M (high concentration) injection, suggesting that they did perceive and react to the odorant stimulus. However, these rare individual responses were "diluted" at population level by the pooling of all fish in the distribution graphs. In summary, SF behavioral responses were modest and markedly different from CF.

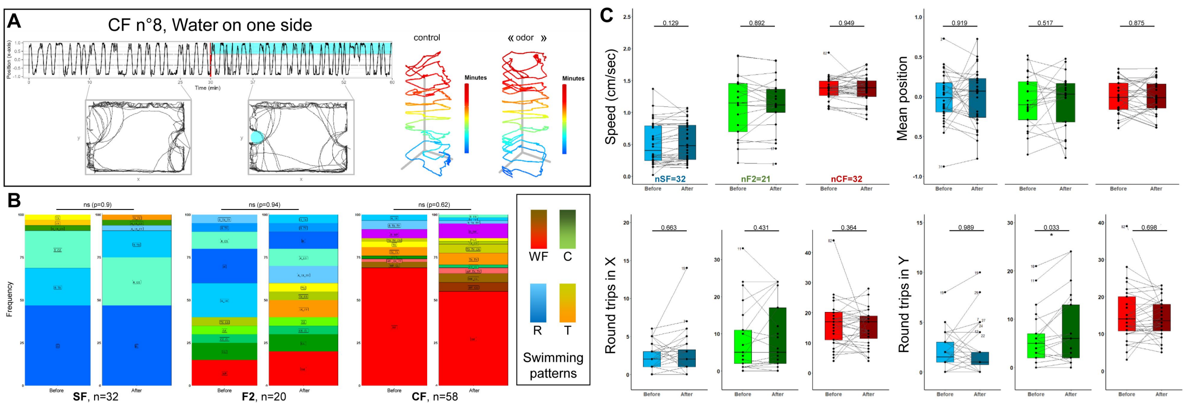


Figure 3

Controls on the olfactory setup and solution delivery method.

A, Lack of behavioral response in a representative CF individual after injection of water on one side of the testbox.
B, Lack of change in qualitative swimming pattern displayed by SF, CF and F2, before and after water injection on one side. Fisher's exact tests.
C, Lack of change in quantitative swimming kinetics in SF (blue), CF (red) and F2 (green), before (pale colors) and after (dark colors) water injection on one side. In this and the following graphs, boxplots show the swimming speed, the fish position along the X-axis of the testbox, and the numbers of back-and-forth swims along the X (length) and Y (width) axes of the box. All values are averaged over a 15 minutes period, either before or after the injection. A thin line links the value « before » and the value « after », for each individual. Numbers of fish tested are indicated on the "speed" boxplots. Numbers next to dots indicate the identity of outlier individuals.

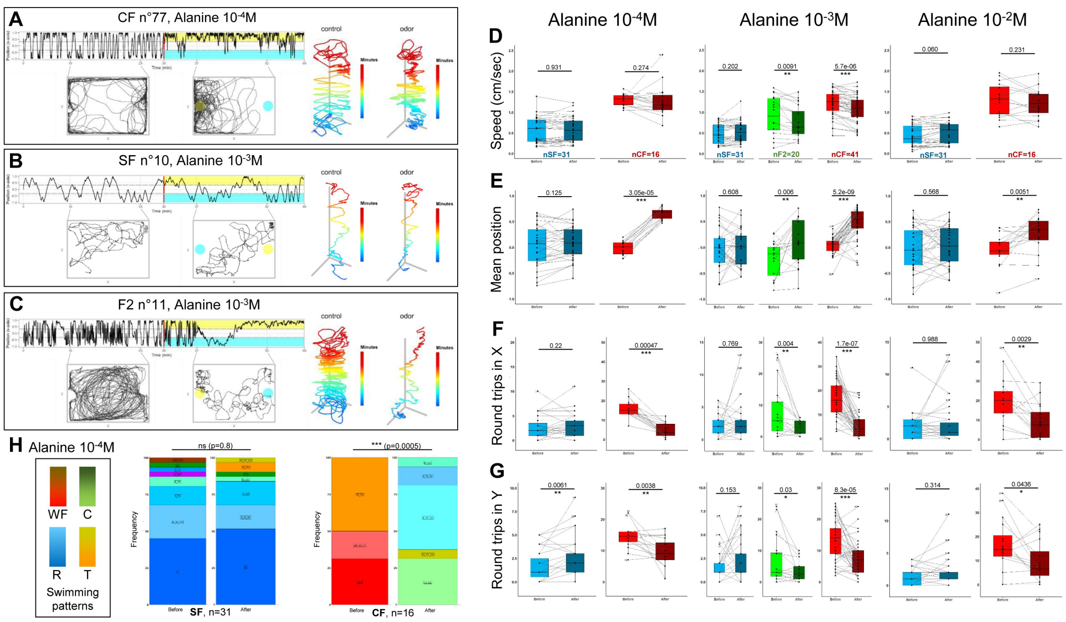


Figure 4

Behavioral responses to alanine of 6 week-old CF, SF and F2.

A,B,C: Representative individual responses of CF, SF and F2 after injection of alanine at the indicated concentrations. In the two left graphs of each panel, the blue color indicates the water/control injection side and the yellow color indicates the alanine injection side.

D,E,F,G: Box plots showing swimming speed (D), mean position (E) and the number of back and forth trips in X and Y (F,G) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) the injection of alanine at the indicated concentration. Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value for each individual. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated.

E, Change in swimming patterns exhibited by SF and CF after injection of alanine 10^{-4}M. Fisher's exact tests.

Finally, F2 were tested with alanine 10^{-3}M, a concentration that elicits strong responses in CF but not in SF. Upon stimulus, F2's change in behaviors were similar to CF (**Figure 4C**; Suppl. Fig2C for other examples): they decreased back-and-forth swimming activity, decreased swimming speed, and swam close to the odor source (**Figure 4DEFG**). They also shifted their swimming patterns (Suppl. Fig2D; Fisher's exact test p=0.0065), including a loss of wall following mode that was reminiscent of the trend observed in CF. As an illustration, the individual shown in **Figure 4C** displays a striking change from a "large circles" to a "random" swimming pattern accompanied with variations in swimming kinematics.

Overall, these data suggested that, compared to SF, CF detection threshold and behavioral responses to the amino acid alanine have significantly evolved.

Behavioral responses to serine and cysteine in *Astyanax* morphotypes

We next systematically tested behavioral responses to injections of 10^{-2}M, 10^{-3}M or 10^{-4}M of serine (polar amino acid, hydroxyl group) and cysteine (non-polar, sulfur containing), two other potent amino acid olfactory cues for fish (**Figure 5** and Suppl. Fig3).

Serine elicited behavioral responses similar to alanine: CF (as well as F2 hybrids, Suppl. Fig3) moved towards the odor source, whereas SF did not (**Figure 5AB** and **D**). Swimming speed was unchanged in all fish types (**Figure 5C**), but a decrease or an increase of back-and-forth swimming activity was observed in CF and SF, respectively (**Figure 5EF**). At population level, significant changes in swimming patterns were observed for the three morphs upon 10^{-2}M serine (**Figure 5G**; Fisher's exact p-values 0.0005, 0.001 and 0.0085 for SF, CF and F2, respectively). Of note, and contrarily to alanine, the lower concentrations of serine tested (10^{-4}M and 10^{-3}M in syringe; 10^{-6}M and 10^{-5}M in box, respectively) were unable to trigger a robust behavioral change in cavefish (not shown, see Table1), suggesting that their detection threshold for serine is lower than for alanine. Regarding SF, responses were also observed only for 10^{-2}M serine injections.

Cysteine provoked moderate effects, even at the highest concentration injected (10^{-2}M) (**Figure 5H-N** and Suppl. Fig3). Swimming speed was unchanged in all morphs (**Figure 5J**). CF were attracted to the odorant side for the 3 concentrations tested (**Figure 5K**) and SF increased the numbers of back-and-forth swims along the Y-axis when responding to 10^{-2}M cysteine (**Figure 5M**). The three fish types changed significantly their swimming patterns (**Figure 5N**; Fisher's exact p-values 0.03, 0.017 and 0.04 for SF, CF and F2, respectively). After injection of cysteine at the lower concentrations of 10^{-3}M or 10^{-4}M, CF behavioral responses were similar to those observed for alanine and serine, i.e., a decrease in back-and-forth swimming activity along the X-axis and a significant change in position towards the odor source, without changing swimming speed (not shown, see Table1).

In sum, the responses of the different fish types to different concentrations of different amino acids were diverse and may reflect complex, case-by-case, behavioral outputs.

Behavioral responses to lysine, histidine and leucine in *Astyanax* morphotypes

Finally, we choose to examine responses triggered by high concentrations (10^{-2}M) of three other, less studied amino acids: lysine, histidine (both polar and positively charged) and leucine (aliphatic, like alanine) (**Figure 6** and Suppl. Fig4).

For these three odors, CF responses were conspicuous and could include a shift of swim position towards the odor's source, a decrease of back-and-forth swimming activity with decrease in swimming speed, and significant changes in swim patterns at population level (except for leucine)

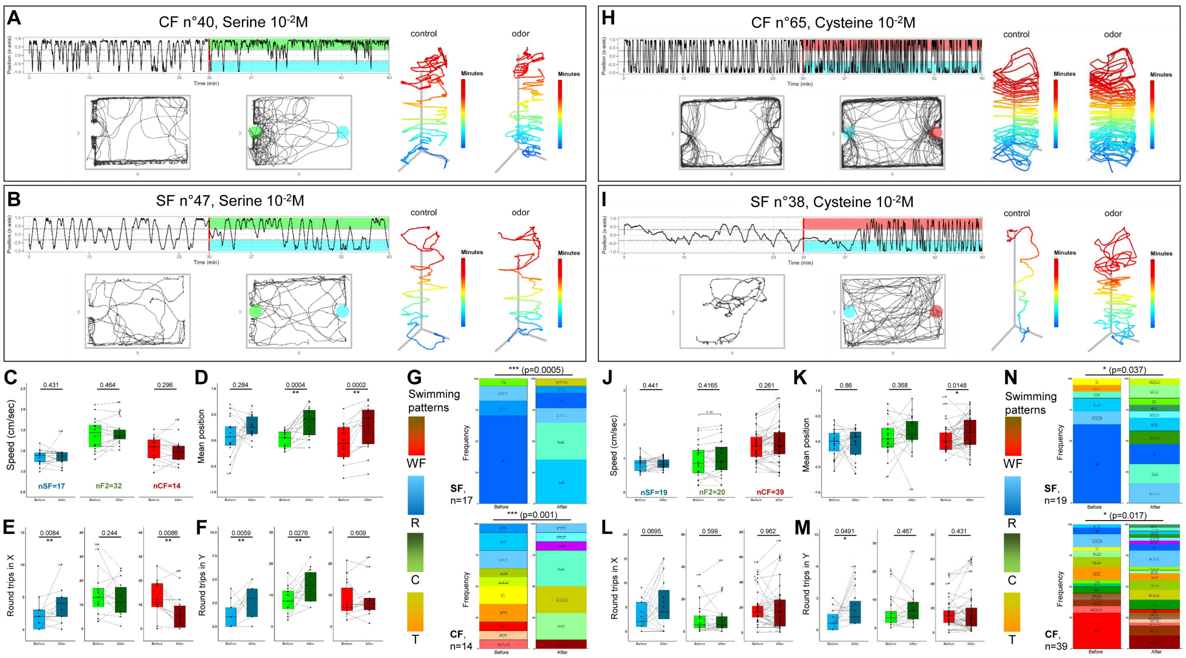


Figure 5

Behavioral responses to serine and cysteine 10^{-2}M.

A,B and **H,I**, Representative individual responses of CF and SF after injection of serine (A,B) or cysteine (H,I) at high concentration (10^{-2}M). In the two left graphs, the blue color indicates the water/control injection side and the green (serine) or red (cysteine) color indicates the injection side.

C-F and **J-M**, Box plots showing swimming speed (C,J), mean position (D,K) and the number of back and forth trips in X and Y (EL and FM) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) injection of serine 10^{-2}M (C-F) or cysteine 10^{-2}M (J-M). Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value of each individual fish. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated.

G and **N**, Change in swimming patterns elicited after injection of 10^{-2}M serine (G) or cysteine (N) in SF and CF (F2 not shown). Fisher's exact tests.

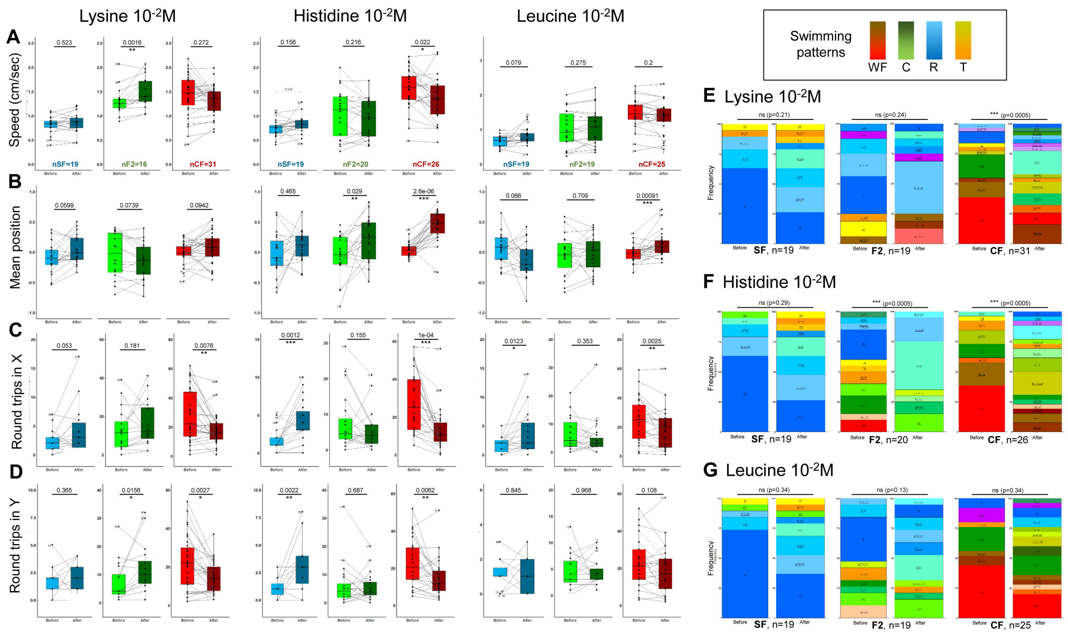


Figure 6

Behavioral responses to lysine, histine and leucine 10^{-2}M.

A-D, Box plots showing swimming speed (A), mean position (B) and the number of back and forth trips in X and Y (CD) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) the injection of the indicated amino acid. Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value of each individual fish. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated. See **Supplemental Figure 4** for examples of representative individual responses. **E-G**, Change in swimming patterns elicited after injection of 10^{-2}M lysine (E) or histidine (F) or leucine (G) in SF, CF and F2. Fisher's exact tests.

(**Figure 6A-G** and Suppl. Fig4). Regarding SF, changes were restricted to increases in back and forth swimming activity for histidine, without change in swimming speed or position in the box (**Figure 6A-D** and Suppl. Fig4). These three amino acids did not elicit changes in swimming pattern in SF at population level (**Figure 6E-G**). Finally, F2 also showed specific qualitative (swimming patterns) and quantitative responses to each of these three amino acids. Together, these data suggested that CF, SF and F2 detected and responded in their own and specific way to high concentrations of lysine, histidine and leucine.

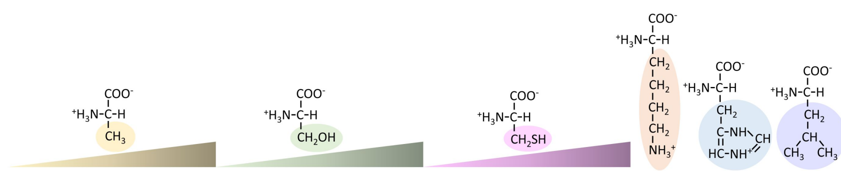
Table 1A provides a summary of the behavioral responses, averaged at population level, for the three fish types and the six amino acids tested. In the cases when significant behavioral responses were observed at population level, i.e., when responses are strongest and most stereotyped, different and opposed features emerged between CF and SF (and F2). As a "rule", CF responses included a decrease in swimming activity (decrease in speed and back and forth swims along X- and Y-axes) together with an attraction to odor source and a change in swim pattern (**Table 1A**, red arrows). Conversely, SF responses mostly corresponded to an increase in swimming activity, with little or no attraction to odor and moderate change in swim pattern (**Table 1A**, blue arrows). In sum, a given sensory stimulus could trigger markedly different, morph-specific behavioral responses.

Behavioral responses to odors at individual level

Analyses as above performed at population level may mask or blur effects or phenotypes in the case when not all individuals respond in a stereotyped manner. As fish did express individual behavioral features in our experimental paradigm, we sought to perform further analyses at individual level, and to calculate individual response scores to the different odors. To do so, and to take into account the different components of the behavioral response, we summed indexes of speed, back and forth trips in X and Y, position and pattern changes (see Methods; **Figure 7A**). Using individual scores of fish in control conditions after perfusion of water, we set the threshold of score for which a fish was considered to respond at 1.5 (Suppl. Fig5). Visual inspection of the responses to odors of fish who had an individual score just above or just below this threshold confirmed that it was accurate.

For alanine 10^{-3}M, the representation of individual behavioral responses shows that CF responded in a very stereotyped manner (all lines following the same "curve") whereas SF responses were more diverse (lines crossing) (**Figure 7A**, 1st column). Moreover, 73% of CF but only 19% of SF had an individual score above threshold - a marked difference that was visible on the distribution of scores (**Figure 7AB**). F2 fish on the other end had a bimodal distribution of their olfactory scores for alanine 10^{-3}M (**Figure 7B**).

For other odors, the difference in olfactory scores of individual CF and SF was less obvious (**Figure 7B** and Suppl. Fig5), suggesting that taking into consideration individual's variations of swimming behavior can unmask responses that are not visible when averaged at population level. For example, in response to cysteine 10^{-2}M, SF (68% with score >1.5) and F2 (55% with score >1.5) had globally better scores than CF (39% with score >1.5) (**Figure 7AB**, 3rd column). Of note, these results had not shown up in **Table 1A** summarizing responses averaged at population level. Noteworthy, SF had mixed, diverse responses to all odors studied (lines crossing on all graphs). Conversely, CF showed non-homogenous responses to serine, histidine, leucine and lysine but highly stereotyped responses to both alanine and histidine (**Figure 7A** and Suppl. Fig5). A summary of individual response scores to the different odors is given in **Table 1B**.



		alanine 10^{-4}M	alanine 10^{-3}M	alanine 10^{-2}M	serine 10^{-4}M	serine 10^{-3}M	serine 10^{-2}M	cysteine 10^{-4}M	cysteine 10^{-3}M	cysteine 10^{-2}M	lysine 10^{-2}M	histidine 10^{-2}M	leucine 10^{-2}M
A population	**Speed**												
	SF	ns	ns	ns	-	↗	ns	-	-	ns	ns	ns	ns
	F2	-	↘	-	-	ns	ns	-	-	ns	↗	ns	ns
	CF	ns	↘	ns	ns	ns	ns	ns	ns	ns	ns	↘	ns
	Position												
	SF	ns	ns	ns	-	ns	ns	-	-	ns	ns	ns	ns
	F2	-	↗	-	-	↗	↗	-	-	ns	ns	↗	ns
	CF	↗	↗	↗	ns	↗	↗	↗	↗	↗	ns	↗	↗
	NbrtX												
	SF	ns	ns	ns	-	ns	↗	-	-	ns	ns	↗	↗
	F2	-	↘	-	-	ns	ns	-	-	ns	ns	ns	ns
	CF	↘	↘	↘	ns	↘	↘	ns	↘	ns	↘	↘	↘
	NbrtY												
	SF	↗	ns	ns	-	ns	↗	-	-	↗	ns	↗	ns
	F2	-	↘	-	-	ns	↗	-	-	ns	↗	ns	ns
	CF	↘	↘	↘	ns	ns	ns	ns	ns	ns	↘	↘	ns
	Patterns												
	SF	ns	ns	ns	-	ns	change	-	-	change	ns	ns	ns
	F2	-	change	-	-	ns	change	-	-	change	ns	change	ns
	CF	change	change	change	ns	ns	change	ns	change	change	change	change	ns
B individual	**Score > 1.5**												
	SF	3.2%	19.3%	42%		25%	41.1%			68.4%	21%	42.1%	21%
	F2		50%			31.3%	18.8%			55%	18.7%	55%	36.9%
	CF	81.3%	73.2%	56.3%	18.7%	6.2%	35.7%	31%	43.8%	39.5%	48.4%	69.2%	28%

Table 1.

Summary of the results.

A, Population level summary. For each amino acid and each concentration tested, arrows indicate whether the considered parameter has changed (increased or decreased). ns indicates no significant change, - indicates the condition was not tested in the fish type (SF in blue, CF in red, F2 in green).

B, Individual level summary. The percentage of fish displaying an individual olfactory score superior to 1.5 is indicated.

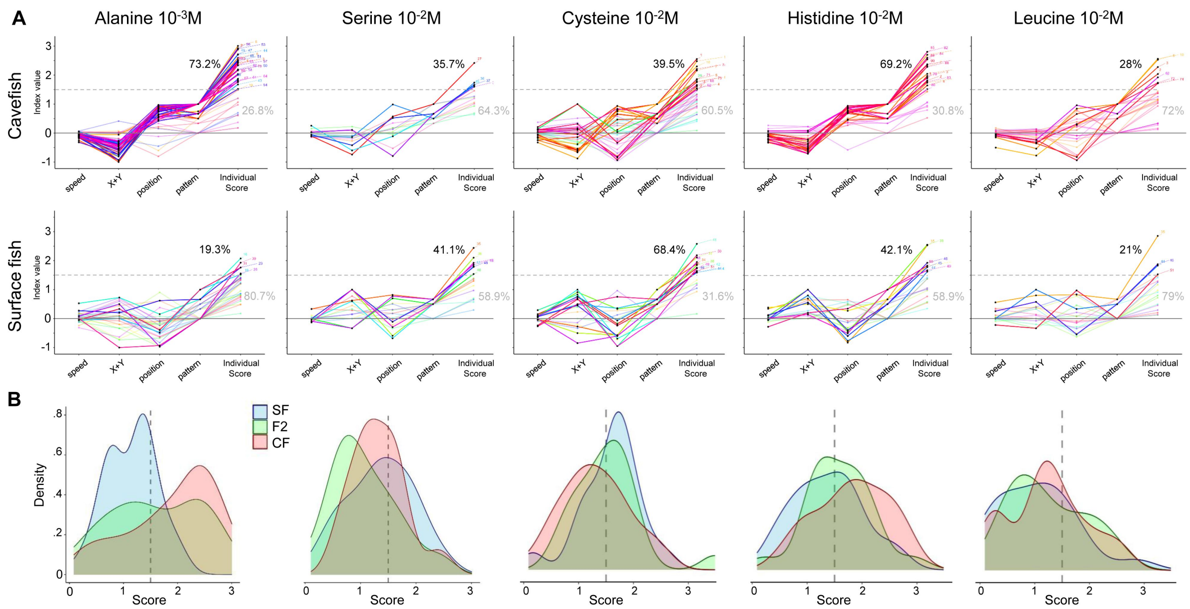


Figure 7

Individual olfactory scores of SF, CF and F2 for different odors.

A, Graphs representing index values (i.e., the variation between the 'before' and the 'after' odor condition; the value 0 corresponds to no change) for the 4 response parameters (speed, thigmotactism in X and Y, position and swim pattern) used to calculate the total individual olfactory score for each fish (last column of points on each graph). Each fish is depicted by a colored line linking its 4 different indexes and its final individual olfactory score. The dotted line at value 1.5 indicates the score threshold above which a fish is considered as a responder. The percentages in black and grey indicate the proportion of responders/non-responders, respectively. The colored lines of responders are bright, those of non-responders are pale. Amino acids and concentrations are indicated. Top row: CF; bottom row: SF.
B, Distributions of individual olfactory scores of SF (blue), CF (red) and F2 (green) for different odors. The threshold score (1.5) is indicated by a dotted line.

Individual personality and individual behavioral response

Finally, the results presented above led us to test whether the “personality” of each fish, represented by their specific baseline swimming behavior, could influence their ability to respond to odors. To this end, we plotted individual olfactory scores as a function of individual baseline swimming patterns (R, WF, TX, TY, or C). We also examined possible correlations between basal swimming speed or round trip activity and individual olfactory score.

In SF, individual olfactory scores were similar for the four basal swimming patterns (**Figure 8A**). However, those fish exhibiting lower baseline swimming speed and less round trips in X had better olfactory scores for several odors (**Figure 8BC**, first column; negative correlation). By contrast, in CF, fish exhibiting WF or C as baseline pattern had significantly better individual scores and those swimming randomly (R) had poor scores (**Figure 8A**) – but scores were not correlated whatsoever to baseline swimming speed (**Figure 8BC**, last column). Moreover, for alanine 10^{-3}M and histidine 10^{-2}M, the two odors for which more than 70% of CF showed a significant response (**Figure 7A**), a high number of round trips in X in baseline behavior was also associated to good olfactory scores (in agreement with WF being predictive of significant individual response). Finally, F2 that exhibited a WF baseline pattern had better scores than others (like CF), and swimming speed was poorly correlated to their olfactory performances. Altogether, these analyses show that (1) the fish swimming personality has an influence on its response to olfactory stimulation, and (2) the personality parameter that is important to predict a good individual response is not the same in the two *Astyanax* morphs: in SF speed matters, while in CF swimming pattern matters.

Discussion

We have developed a high throughput, specific olfaction test, together with a pipeline of analysis allowing assessing qualitatively and quantitatively individual’s and population’s behaviors, in order to describe and compare responses of blind and sighted *Astyanax* to odorant cues. We discovered that cavefish, surface fish and F2 hybrids display different odor preferences and sensitivities and show individual, distinctive, diverse and specific responses to varying concentrations of the six amino acids tested. Using this novel setup where fish were tested in solo in a rectangular box and during one hour (as opposed to testing in groups in a U-shaped box and during 8 minutes in previous studies (Blin et al., 2020; Blin et al., 2018; Hinaux et al., 2016a), we established that cavefish are *bona fide* “alanine specialists” and we analyzed in depth their behavioral responses.

A setup to probe olfaction in fish with markedly different sensory apparatuses and internal states

There is an inherent difficulty to compare sensory-driven behaviors in cave and surface morphs of *Astyanax*: all their sensory systems have evolved in one way or another. Cavefish have no eyes, but they possess enhanced mecanosensory lateral line and chemosensory gustatory and olfactory organs. To decipher behavioral responses driven by a single of these senses, one needs to control carefully the potential influence of the other sensory modalities. Here, we have recorded unimodal behavioral responses driven exclusively by olfaction by performing experiments in the dark to abrogate vision in sighted fish, and by designing a setup where the delivery of the olfactory stimulus is vibration-free. Moreover, we know from previous studies that gustatory taste buds do not participate in responses to amino acids at the concentrations used because the lesion of the olfactory epithelium abolishes the attraction to high concentrations of alanine, in both SF and CF (Hinaux et al., 2016a).

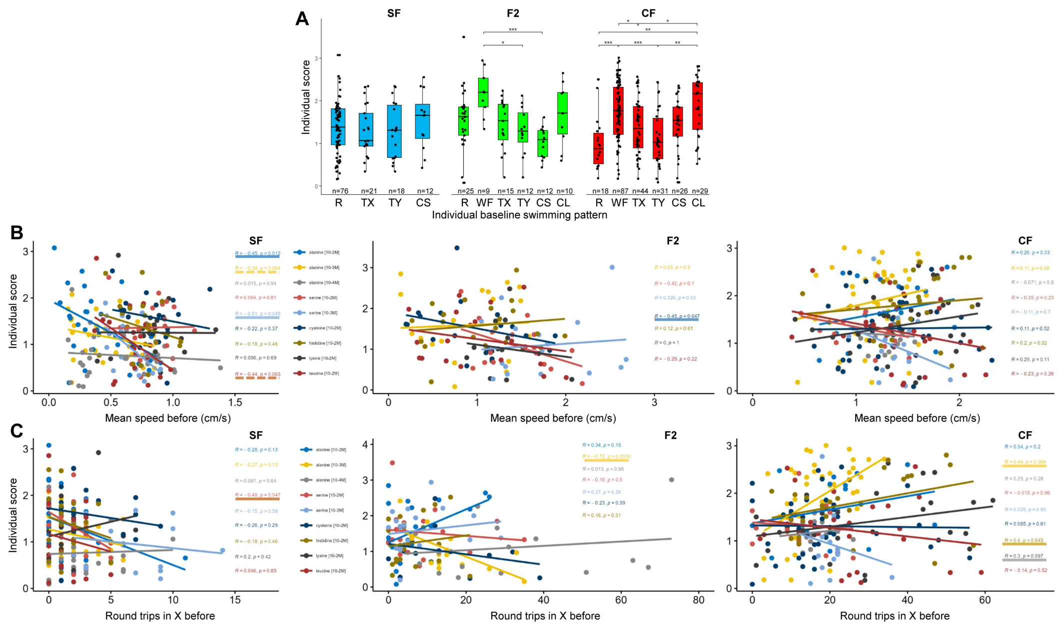


Figure 8

The individual olfactory score is related to individual fish personality.

A, Box plots showing individual olfactory scores in SF (blue), F2 (green) and CF (red) as a function of their individual baseline swimming pattern. As stated in Methods, only amino acid conditions for which more than 40% of fish were responders (sore >1.5) are pooled and plotted on this graph. Mann-Whitney two-tailed with Bonferroni post hoc tests were performed on each morphotype.

B,C, Regressions to explore relationships between baseline swimming speed (B) and number of round trips in X (C) displayed by individual fish before odor injection and their individual olfactory score. Fish and amino acid type and concentrations are indicated. Linear correlations were calculated with Spearman's rank correlation test followed by Student test for the p values. Conditions for which significant correlation was found are underlined (continuous line for p<0.05; broken line for p=0.06).

Due to a mutation in their Monoamine Oxidase enzyme that interferes with the metabolism of brain monoamines (Elipot et al., 2014), Pachón cavefish have lower basal cortisol levels, hence lower basal anxiety than surface fish, when they are long habituated in their home tank (Pierre et al., 2020). However, the mutation confers cavefish with a much higher stressability after environmental change, such as the transfer in a novel tank (Pierre et al., 2020). Consequently, in order to record relevant behavioral responses to olfactory stimuli in unstressed fish, we used long acclimation times (72 hours) and we tested long habituation times, either 1 hour or 24 hours – as compared to 10 minutes as usually done in most fish studies including zebrafish or *Astyanax*. We concluded that 1 hour is too short for proper habituation, as the swimming speed and patterns were affected for both SF, CF and F2 as compared to 24 hours. Importantly, such long, 24 hours habituation periods are susceptible to reveal unperturbed and even novel behaviors: recently applied in a study of social behaviors in cavefish, 3 days of habituation allowed the analysis of behaviors in a familiar environment and could unmask social interactions in the so-called 'asocial' cavefish (Iwashita and Yoshizawa, 2021).

Neurophysiological and molecular considerations

In all vertebrates including fish, odorant molecules are recognized by olfactory receptors expressed at the surface of olfactory sensory neurons, which project onto the olfactory glomeruli in the olfactory bulb with the one receptor : one glomerulus rule (Axel, 1995; Braubach et al., 2012; Buck, 2000; Kermen et al., 2013; Koide et al., 2009; Li et al., 2005; Yoshihara, 2008). Moreover, parallel neural pathways in the olfactory circuitry process different types of odorants. In zebrafish, the perception of amino acids is mediated via OlfC/V2R receptors on microvillous sensory neurons that innervate lateral and ventro-medial glomeruli in the bulb. From the periphery to the brain, *Astyanax* surface and cave morphs display some variations in this amino acid signal processing circuitry. SF have 43 and CF have 41 V2R/OlfC receptor genes in their genomes (Policarpo et al., 2022), a minor difference that is unlikely to underlie their differences in olfactory capacities and preferences. CF larvae have higher proportions of microvillous neurons than SF, which together with the larger size of their olfactory epithelium and olfactory bulbs may influence their olfactory sensibility (Blin et al., 2018). *Astyanax* glomerular organization is unknown.

Interestingly, here we have found that CF are strongly attracted and respond to alanine and histidine, two amino acids which, albeit probably not recognized by the same receptor(s), are processed in the same or very close glomeruli in zebrafish larval olfactory bulbs (Li et al., 2005). SF on the other hand seem to show a preference for cysteine (sulfur), the amino acid that is the most potent to evoke electrical responses in the olfactory bulbs of sea breams (Hubbard et al., 2011) or hammerhead sharks (Tricas et al., 2009), and that elicits a strong aversive behavioral response in larval zebrafish (Vitebsky et al., 2005). Therefore, odor preferences have evolved between cavefish and surface fish, as well as between zebrafish and *Astyanax*. In the same line, previously we had reported that chondroitin is a strong attractant for *Astyanax* (Blin et al., 2020), whereas it induces freezing and fear behavior in zebrafish (Mathuru et al., 2012). Such significant variations in odor preferences or value may be adaptive and relate to the differences in the environmental and ecological conditions in which these different animals live. However, the reason why Pachón cavefish have become "alanine specialists" remains a mystery and prompts analysis of the chemical ecology of their natural habitat. Of note, we have not found an odor that would be repulsive for *Astyanax* so far, and this may relate to their opportunist, omnivorous and detritivore regime (Espinasa et al., 2017; Marandel et al., 2020).

Cavefish have also evolved regarding olfactory sensitivity. They are able to detect very low concentrations of alanine (aliphatic), hence they have a lower detection threshold than SF. Injection of low concentration alanine 10^{-4}M (thus 10^{-6}M in the odorant third of the arena) elicits strong behavioral responses in CF, whereas even higher concentrations of 10^{-3}M and 10^{-2}M evoke modest responses in SF (zero response at population level, 19% responders at individual level with alanine 10^{-3}M). Although we have not performed dose-response experiments for all amino acids tested and for both morphs, CF also appear capable of detecting low concentrations of cysteine (sulfur). Moreover, they seem to detect better histidine 10^{-2}M (polar) as well (69.2% of CF responders, versus 42.1% of SF), and CF but not SF detected lysine (polar) at the relatively high concentration of 10^{-2}M. From these observations, we can predict that the difference between SF and CF probably does not lie in the molecular evolution of their olfactory receptor repertoire, because it is unlikely that all the receptors recognizing diverse types of amino acids have evolved all at once. Rather, we can hypothesize that evolution occurred at the level of the general regulation of odorant receptor genes expression, or at the level of olfactory processing and computation in the bulbs (Friedrich et al., 2009), or in higher order brain regions.

A variety of behavioral responses to olfactory stimulation

Our study suggests that behavioral responses to a unimodal olfactory sensory stimulus are complex and have evolved in cavefish (summary on **Table 1**). When they detect the odor stimulus, cavefish globally decrease their scanning activity, which might help them to compute and swim up towards the higher values of concentration in the odor plume. This is associated to a change in swimming pattern whereby WF and T are eliminated at the expense of R and C, presumably to facilitate searching. This hypothesis is further supported by their systematic change of position in the box, meaning that they locate efficiently the odor source. Strikingly, surface fish display opposite behavioral responses after odor detection: they rather increase locomotor activity, which corresponds to intense foraging but does not seem optimal to find the odor source, which is confirmed by their lack of change in position of the box (i.e., they may not locate efficiently the odor source). This is also consistent with the fact that at individual level the best SF responders are slow swimmers. Such poor ability to find the odor source may result from testing in the dark. Indeed, SF behaviors are mostly visually driven (but see (Simon et al., 2019)) and they might need multimodal visual + olfactory integration to find food efficiently. These interpretations are consistent with early work showing that, when competing in the dark and with limited amounts of food, SF starve and CF thrive (Hüppop, 1987). In sum, CF foraging strategy has evolved in response to the serious challenge of finding food in the dark. Future experiments including functional imaging of brain activity in live animals may reveal the changes in olfactory-driven motor circuits that allowed the evolution of behavioral outputs in cavefish.

Finally, at population level, F2 fish shared some behavioral response traits with both parental morphs, and they were often closer to CF. At individual level their responses were also variable and non-stereotyped (including for alanine 10^{-3}M and histidine 10^{-2}M, the conditions for which CF showed highly stereotyped responses). Olfactory scores behave as a quantitative trait. The tools we have developed here will allow the future determination of the genetic underpinnings of the evolution of olfactory-driven cavefish behaviors and capabilities.

A “personality” for each fish?

Our recordings of several hundred (n=425 total) well-habituated, individual *Astyanax* larvae highlight an often overlooked aspect of fish behavioral analyses: fish may have a “personality” or “temperament”, which we characterized at the level of their baseline swimming patterns. This hypothesis is strongly supported by the consistency over time, one day after the other, of the favorite swimming pattern or combination of patterns expressed by individuals when freely swimming without stimulus (most fish were recorded on 4 days distributed over 2 weeks). To our knowledge, this is the first time individual variations are taken into consideration in *Astyanax* behavioral studies.

In the context of ecology and evolution, the personality in non-human animals was proposed to be organized along five primary axes: sociability, boldness, aggressiveness, exploration and activity (Réale et al., 2007). In fish, studies applying an animal personality approach have focused to resolve variations in physiological and molecular parameters, suggesting a link between phenotype and genotype, or between behavior and transcriptome regulation eg (Rey et al., 2021). Here, we propose that the different temperaments expressed by individual larvae could correspond to the genetic diversity present in the natural populations of *Astyanax*, a diversity that we have maintained intentionally along generations in captivity in our fish facility. Further, we have assessed whether swimming temperaments could influence the way and the extent to which fish respond to an olfactory sensory stimulus. Strikingly, we discovered that baseline swimming patterns and swimming speed do influence fish olfactory responses. In addition, it appears that important personality traits to confer positive behavioral responses are not the same in SF and CF. For cavefish, WF and round trips in X (but not speed) are the key parameters. Wall following behavior has been reported in *Astyanax* and other blind cavefish species (Chen et al., 2022; Patton et al., 2010; Sharma et al., 2009) and was hypothesized to confer foraging advantages, although this had never been directly tested. Here, it seems to be the case for 6-week old juveniles in a rectangular box - but the link may be more elusive when considering a fish swimming in a natural, complex environment. We propose that WF and round trips in X are together the expression of an intense exploratory behavior, which allows CF to better cover their swimming space and thus have a higher probability to detect odorant cues - even at very low concentrations because their detection capacities for some amino acids are excellent. By contrast, a slow swimming speed (but not swimming pattern) is the most critical personality parameter for SF. As SF olfactory concentration detection thresholds are higher than CF, one could imagine that they need more integration time inside the odorant plume to trigger a neuronal and a behavioral response. In any case and importantly, personality parameters that matter in CF and SF to confer good olfactory response scores are distinct. This suggests that the modulation of neuronal circuits underlying both the control of baseline behaviors and olfactory processing has evolved in cavefish.

Conclusion

Previous studies had shown that, when tested in groups, cavefish larvae as well as adults smell better than their surface fish conspecifics (Blin et al., 2020; Blin et al., 2018; Hinaux et al., 2016a). Here, we have established that it is also true for fish in solo, ruling out the possibility that when in group, one "good smeller" individual could drive others to respond through (unknown) communication mode. Using individual behavioral tests, we have discovered that both olfactory sensitivity, olfactory preferences, olfactory behavioral responses and key personality parameters have evolved in cavefish, conferring them with outstanding skills to forage in the darkness of caves.

Methods

Fish samples

Laboratory stocks of *A. mexicanus* surface fish (SF, origin: San Salomon spring, Texas, USA) and cavefish (CF, Pachón population) were obtained in 2004 from the Jeffery laboratory at the University of Maryland, College Park, MD, USA. Fish were maintained at 22°C with a 12h:12h light:dark cycle, fed twice daily with dry and live food. SF and CF brood stock were induced to spawn once every two weeks by thermal shock at 26°C. A F1 hybrid family was generated by crossing a SF female with a CF Pachón male. Second-generation hybrids (F2) were generated by crossing two F1 individuals. Embryos and larvae were reared at 24°C in water produced by our animal house's water production system. All fish used (CF, SF and F2) were exactly six weeks old on the first day of the experiment. SR's authorization for use of *Astyanax mexicanus* in research is 91-116. Animals were treated according to the French and European regulations for animal testing in research (authorization n°APAFIS#8604).

Experimental set-up

The test consisted of filming individual larvae for 30 minutes before olfactory stimulation and for a further 30 minutes afterwards. The behavior laboratory is a soundproofed room maintained at a constant temperature of 24° C. To ensure that eyed fish (SFs and F2 with eyes) could not use the visual modality during the test (unlike CFs and some F2), all experiments (except one) were carried out in the dark. The experimental device was composed of 2 infrared (IR) tables (60×60cm each) over which 32 test boxes were placed. A custom-built wooden holder supported 8 IR cameras (DIGITNOW!), located 40 cm above the boxes. Each camera filmed 4 boxes simultaneously (**Figure 1**).

The individual test boxes (11.5×8.5×4.3 cm, multiroir #45103BOILAB03) were made of crystal-clear plastic. Odorant solutions (and/or water) were manually injected at the two opposite sides of the box using 1ml syringes. In order to stop the dispersion wave of injection, and to assure that we always injected at the same right place, we added 2 injection guides in each box: we glued two plastic tubes (4 cm long and 1 cm in diameter) at the center of the two opposite short sides (Y), 0.2 cm from the bottom. In addition, a net (6.5×6.5 cm) was attached around each tube and reached the bottom of the box, to prevent fish from passing under the tube during the experiment, thus escaping from the subsequent tracking (**Figure 1**).

Behavior tests

Larvae were placed individually in a box containing 150 ml of water (clean rearing water), fed with micro-worms (three drops of concentrated *Panagrellus redivivus* solution placed in the center of the box) and subjected to a 12:12h day:night cycle for 72 hours acclimation. Then, the water in each box was changed, the boxes were placed on IR tables for 24 hours, and the larvae were fasted (=habituation). The day of testing, for reasons of reproducibility, all experiments started at 11 a.m. The light was switched off, the cameras were switched on and the experimenter left the room for a 30-minute baseline behavior recording period. At t=30 min, the experimenter entered the room and switched on the inactinic red lamps. Solutions were manually injected bilaterally (500 µl of odor/ 500 µl of water) into the injection guides of each box. The total time needed to perform injections of 32 boxes was 5 minutes. Then, the experimenter switched off the inactinic lamps and left the room for a 30-minute response behavior recording period. At the end of recording, the water was changed and each box was placed back exactly at the same location on IR table for 24 hours and not moved again until the following day of testing (the larvae were not fed). For the series of experiments with a habituation period of 1 hour, the water in the boxes was changed every morning, 1 hour before the start of the test. The typical timeline was as follows: [72 h acclimation and feeding], Week1 [habituation 24h, test day 1, habituation, test day 2, rest and feeding for three days], Week2 [habituation, test day 3, habituation, test day 4]. As the larvae were tested on several consecutive days, to ensure that the fish were not learning/associating one side with a scent cue, the odor injection side was reversed each day. The control experiments (unilateral or bilateral injection of 500 µl rearing water or no injection) were always carried out the day1 when the larvae were naive. For the “light vs dark” control experiment, SF were filmed 30 minutes in normal light and 30 minutes in the dark.

Odorant solutions

The odorant solutions were prepared by dilution of amino acids in clean rearing water (L-Alanine #A7627; L-Serine #S4500; L-Lysine #L5501; L-Histidine #H8000; L-Leucine #L8000 and L-Cysteine #168149; all from Sigma-Aldrich). A volume of 500µl (either water or odor) was injected with a 1ml syringe (Sigma-Aldrich #Z683531) fitted with a needle (0.8×50mm, 21G; B Braun #4665503). Depending of the test, concentration of solutions injected were 10^{-2} M or 10^{-3} M or 10^{-4} M. In a total water volume of 150ml, with a box virtually separated into three, the volume of water in the odor injection zone is 50 ml. By injecting 500 µl of solution into this zone, we expected to obtain a dilution factor of 100. For example, for a concentration of [10^{-3} M] in the syringe, the concentration in the odor injection zone will therefore be ~[10^{-5} M]. All concentrations given in the article indicate the concentration inside the syringe. When consecutive injections of an odor or different odors at different concentrations were tested on the same fish, they were performed from the least concentrated to the most concentrated.

Video editing and tracking

Videos were saved on SD card in AVI format (1208×720 px, 30 fps), then edited using Adobe Premiere Pro V14.0 to be calibrated at 1 hour long without the 5 minute-injection time. They were then exported into MPG2 format. The tracking software used was TheRealFishTracker V.0.4.0 (*https://www.dgp.toronto.edu/~mccrae/projects/FishTracker/*), developed and freely available from the University of Toronto. In our hands, it was the only software able to detect properly transparent CF larvae. The parameters used were Confidence Threshold = 10; Mean Filter Size = 1; signed Image = *dark object* and all other parameters were the default ones. The x/y-scales were drawn and given in cm. The output data file provided 30 X/Y coordinates (in pixels and in cm) per seconds for a 1-hour movie.

Graphical representations, quantifications and statistics

Statistical analyses were carried out with R-3.4.2 software (R Development Core Team, 2016) using the stats and rstatix libraries, and all graphical representations were designed using the ggplot2, ggrepel, gghighlight and RGraphics libraries.

Quantitative parameters

From the output data set, pixel coordinates Xpi and Ypi were first recoded using the box center as 0, Xcmax = 1 on the odor side and Xcmin = -1 on the water side. The same was applied to the Y coordinate with Ycmax = 0.74 and Ycmin = -0.74. Fish position in the box during test was calculated using Xc and Yc at each time point (30/s). Speed between two time points (5-second time step) was calculated thanks to pythagoras's theorem using X/Y in cm. Round trip number is the total number of time the fish crosses the 0.5 and - 0.5 position (along X-axis) divided by two. Idem for round trips in Y-axis. Quantitative parameter means were all calculated over a 15-minute period, from minute 10 to minute 25 for the 'before' period, and from minute 37 to minute 52 for the 'after' period. Means are represented by box plots showing the distribution of individual values of each fish (black dots), the median, the 25^{th}- 75^{th} percentile and outliers indicated by fish ID number. Paired Mann-Whitney two-tailed tests were performed to compare means of position, speed, or round trips before and after odor injection.

Non-paired Mann-Whitney two-tailed tests were performed to compare means of speed between 1h and 24h habituation periods represented in **Figure 2C** by box plots.

Swimming patterns

The swimming activity of the fish in the test box was systematically represented under 3 different forms, which allow grasping the details and different aspects of the behaviors (see **Figure 2A**). Xc /t coordinates were used with the geom_line() function for the 'Position (X-axis)' graph, (Xc,Yc) /t coordinates were used with the geom_path() function for '2D' top view, and the gg3D package for '3D' view.

The determination of baseline swimming patterns and swimming patterns after odor injection was performed manually based on graphical representations. Swimming patterns in different conditions were compared using Fisher exact test comparing the quantity and distribution of different swimming patterns between SF/CF/F2 at day1, or over day1 to day4, or between 1h *vs* 24 h habituation period, or before *vs* after odor injection for each morphotype.

Index and score

Indexes of position, speed and number of round trips were calculated using the equation (mean *after* – mean *before*) / (mean *after* + mean *before*) commonly used in olfactory tests in fish (Choi et al., 2021; Koide et al., 2009; Wakisaka et al., 2017). An index close to zero indicates that the values of the considered parameter before or after odor injection are close. The higher the absolute value of the index, the greater the difference. When the parameter increases after injection, the index is positive; when the average decreases after injection, the index is negative. The index of pattern change is the number of new patterns observed after stimulation, divided by the total number of patterns observed after odor injection. An index equal to zero indicates no pattern change after stimulation. The higher the index is, the greater the difference of pattern is. For each fish, the individual olfactory score for a given odor and concentration is the sum of the absolute values of the four indexes. On **Figure 7A** and Suppl. Fig5, parameter indexes and fish olfactory scores are represented by dots connected by a colored line for each individual. Those with a score > 1.5 are highlighted and considered to display a significant behavioral response to the stimulus.

On **Figure 8**, to assess the relationship between individual olfactory score and baseline behavior, we pooled results of odors for which more than 40% of individuals have an olfactory score >1.5. So for SF, alanine [10^{-2}M] / cysteine [10^{-2}M] / histidine [10^{-2}M] / serine [10^{-2}M] have been used; for CF, alanine [10^{-2}M]/ alanine [10^{-3}M]/ cysteine [10^{-2}M] / histidine [10^{-2}M] / lysine [10^{-2}M]; and for F2, alanine [10^{-3}M] / cysteine [10^{-2}M] / histidine [10^{-2}M] (see **Figure 7A**). Mann-Whitney two-tailed with Bonferroni post hoc tests were performed on each morphotype.

Slopes in linear regressions show relationships between mean speed or number of round trips before injection and olfactory score. Linear correlations were calculated with Spearman's rank correlation test followed by student test for the p.values.

Supplementary information and data availability

The R script used to recode raw data into corrected coordinates, to calculate means and to draw 2D and 3D graphs is available on GitHub.

F1 and **F2 Files** correspond to source data files (.txt) containing the raw data presented and analyzed in this paper. For each type, the files provide the ID numbers of the fish, condition (day1 to day4 or odor or control type), means of position, speed, X and Y round-trips and behavior for periods 10-25 minutes and 37-52 minutes, as well as the calculated individual olfactory score.

Acknowledgements

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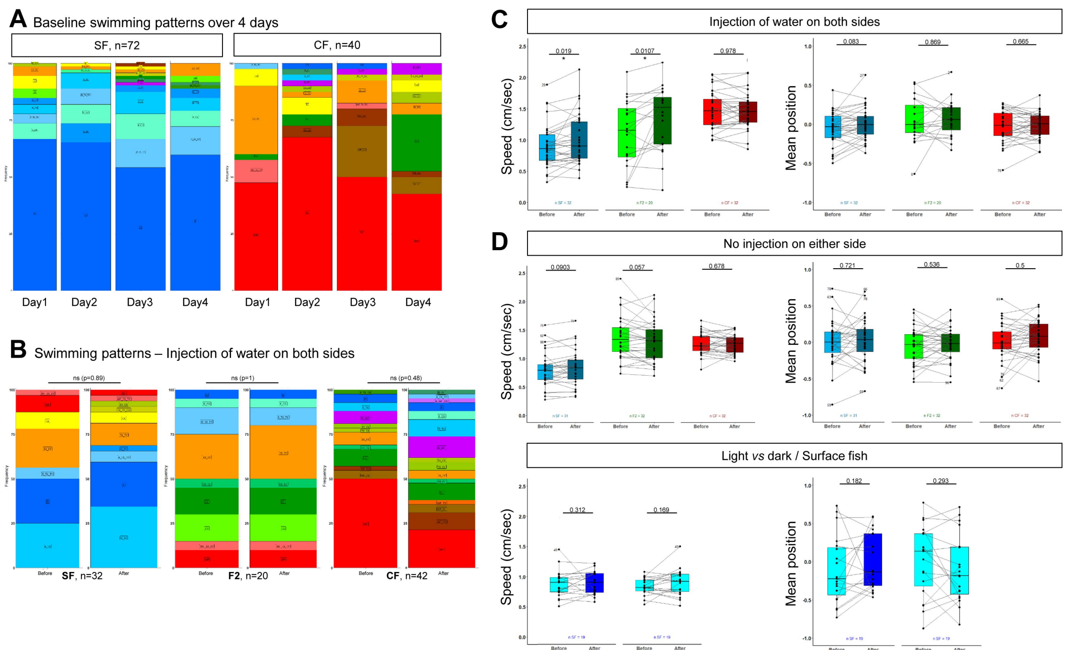
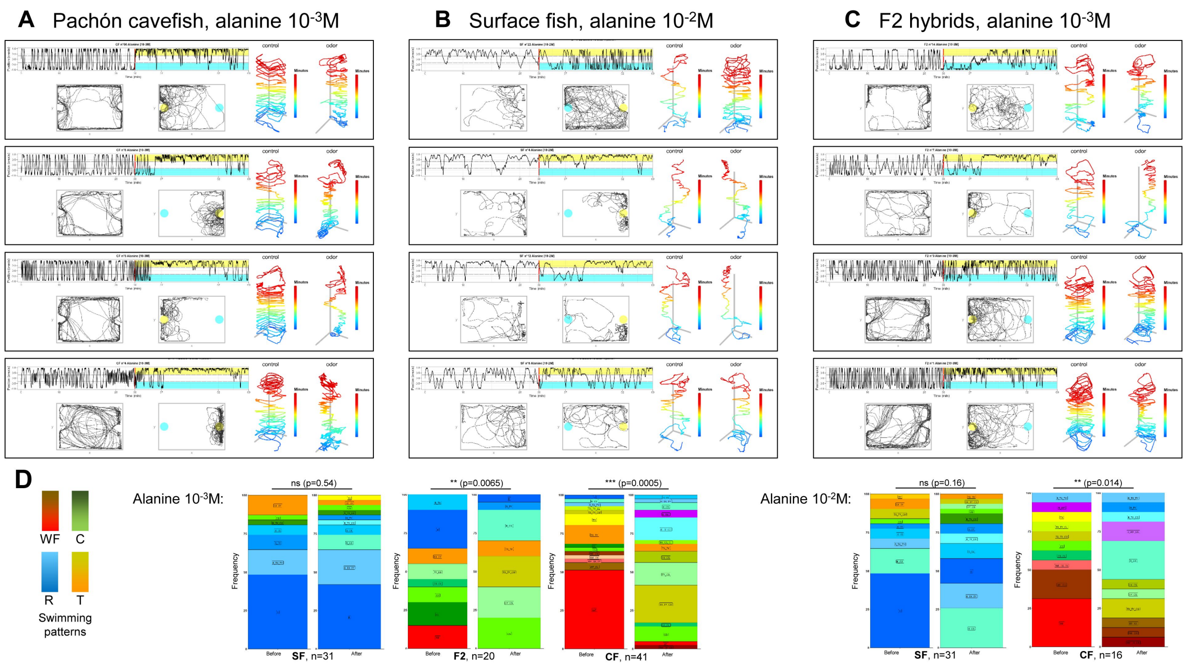
Supplemental Material



Suppl Fig 1

(with Figure 3). Additional controls for experimental design.

A, Distribution of swimming patterns exhibited by SF and CF (F2 not shown) along 4 testing days, with 24 hours habituation time.
B, Lack of change in swimming pattern displayed by SF, CF and F2, before and after water injection on both sides.
C, Global lack of change in quantitative swimming kinetics in SF (blue), F2 (green) and CF (red), before and after water injection on both side. Mean speed and mean position are shown.
D, Lack of change in quantitative swimming kinetics in SF, CF and F2 when no injection is performed at all. Mean speed and mean position are shown. The result suggests that an individual's behavior is stable throughout the one hour test.
E, Lack of change in swimming speed or position on the box when SF are recorded in the dark (dark blue), in the light (light blue), or when the light is switched off after 30 minutes.



Suppl Fig 2

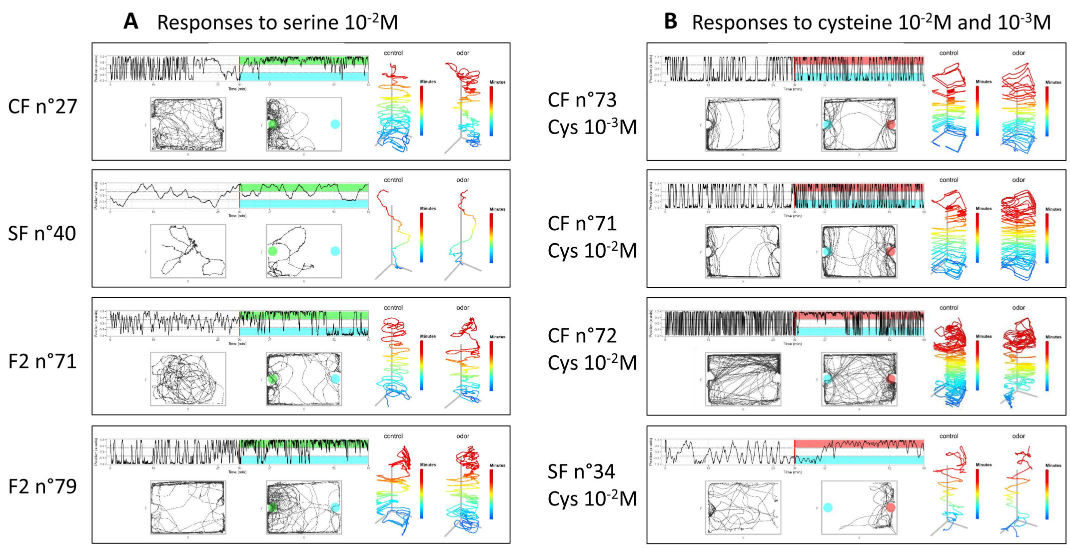
(with Figure 4). Additional examples of behavioral responses to alanine

A, Examples of CF responses to alanine 10^{-3}M (intermediate concentration). Note the reproducible switch in position toward the odorant side in the box (yellow shading) and the change in behavioral pattern. Note that CF n°8, which was tested twice with the same concentration of alanine, switched from wall following to small circles and thigmotactism along the Y border of the box in both cases. CF n°4 on the other hand swam in large circles before injection, and changed its pattern to display intense thigmotactism accompanied with a few circles.

B, Examples of SF responses to alanine 10^{-2}M (high concentration). Upon injection, SF show changes in swimming patterns and increase their swimming speed, suggesting that the stimulus was perceived. SF n°22 swam very actively randomly. SF n°4 switched from random with small circles to circles plus thigmotactism. SF n°12 added thigmotactism to its random pattern upon stimulus. Some SF, like SF n°8, did not seem to respond, even at such high alanine concentration.

C, Examples of F2 hybrids responses to alanine 10^{-3}M (intermediate concentration). F2 n°14 did not appear to move its position to the odorant part of the box, but a striking change in swimming pattern from intense thigmotactism to random swim was observed. F2 n°7 explored close to the odor source and decreased swimming speed. F2 n°3 and 1 switched from wall following to circles with thigmotactism associated to a change in position towards the odor side.

D, Change in swimming patterns displayed by SF, F2 and CF after injection of alanine 10^{-3}M and 10^{-2}M. Fisher's exact tests.

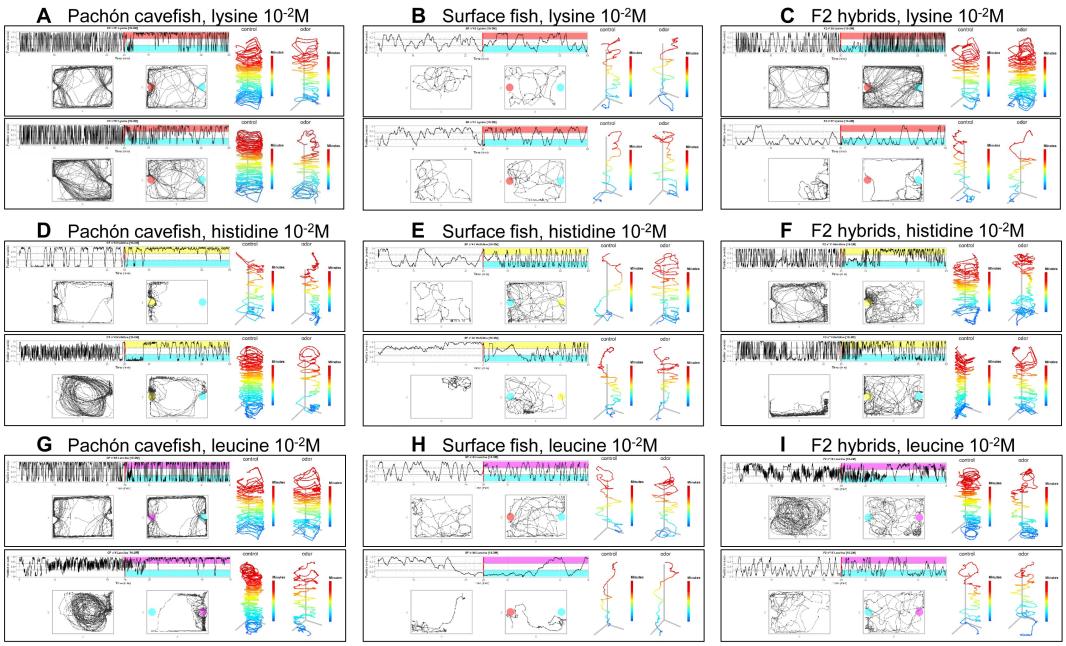


Suppl Fig 3

(with Figure 5). Additional examples of behavioral responses to serine and cysteine

A, responses to serine 10^{-2}M (high concentration). Note the change in behavioral pattern and position in the box observed in CF and F2 but not in SF. The two F2 shown had different baseline swimming patterns: circles in F2 n°71, thigmotactism in F2 n°79. Yet, both individuals showed a change in swim pattern and an attraction to the serine side of the box.

B, responses to cysteine 10^{-2}M (high concentration) and 10^{-3}M (intermediate concentration). CF n°73 gives an example of an individual that did not respond to 10^{-3}M cysteine: no change in position, no change in pattern, no change in swim kinetics. CF n°71 and CF n°72 on the other hand did respond to cysteine 10^{-2}M, the former by an increase in activity, the later by a transient change in position. SF n°34 is a rare example of a SF that was attracted to cysteine.



Suppl Fig 4

(with Figure 6). Examples of behavioral responses to lysine (ABC), histidine (DEF) and leucine (GHI) 10^{-2}M (high concentration)

In each panel, in the two left graphs, the blue color indicates the water/control injection side The lysine injection side is shown in red, the histidine injection side is yellow, the leucine injection side is pink.

A, The 2 CF individuals shown decrease their swimming activity upon injection. Moreover, n°81 and 87 shift their position to the odor side of the box. A change in swimming pattern from intensive wall following to thigmotactism with circles and some random swimming is also observed.

B, The 2 SF individuals shown do not change their swimming pattern (Random) or kinetics in response to lysine.

C, F2 n°35 increased swimming activity upon lysine injection but did not show attraction to odorant side. F2 n° 37 kept its random pattern of swimming and was not attracted to the odorant side.

D, Whether the cavefish individual was initially displaying thigmotactism (CF n°8) or swimming in large circles (CF n°4), the response to histidine included decreasing the swimming activity and restricting locomotion to the odorant side of the box.

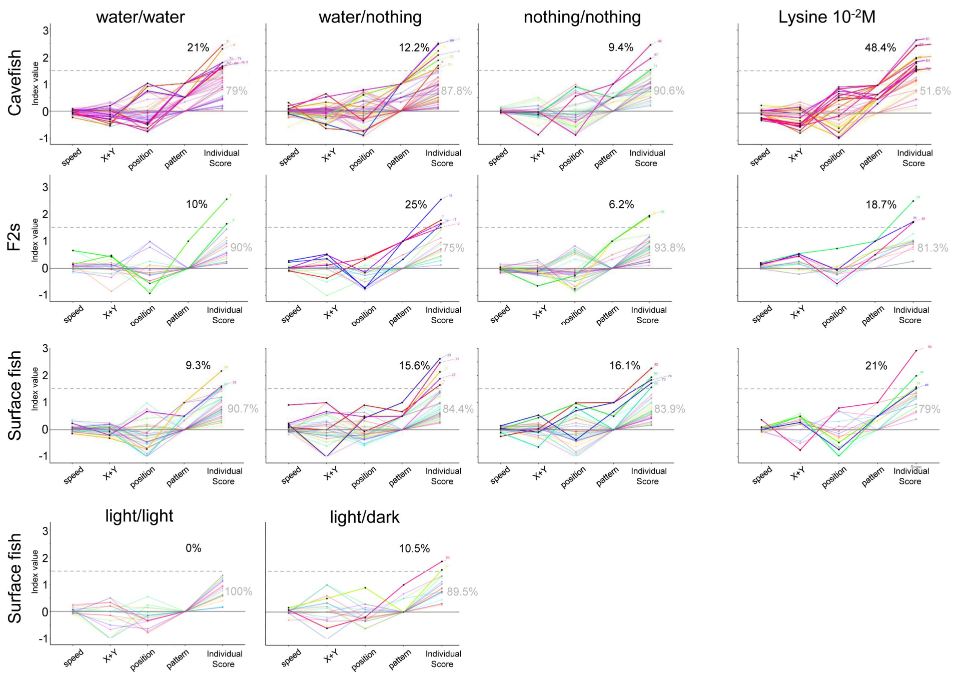
E, The two surface fish individuals shown swam slowly and randomly as baseline pattern. Both increased their activity and spent more time in the odorant side.

F, F2 hybrids n°11 and n°1 were both attracted to histidine odorant side and they changed their swimming pattern.

G, CF n°65 showed no response, neither in activity, in position nor in pattern. CF n°4 drastically changed all swimming parameters.

H, Surface fish showed little reaction to leucine. SF n°43 and n°40 persisted in random swimming patterns, without change of position or swimming kinetics.

I, F2 hybrids n°16 and n°13 had markedly different baseline patterns (large circles *versus* random swim, respectively). The former changed its pattern to less active and small circles, suggesting that it perceived leucine injection. The later showed no behavioral response.



Suppl Fig 5

(with Figure 7). Individual olfactory scores of SF, CF and F2 in control and experimental conditions.

The graphs represent index values for the 4 response parameters used to calculate the total individual olfactory score for each fish. Each fish is depicted by a colored line linking its 4 different indexes and its final individual olfactory score. Experimental conditions and fish types are indicated. The threshold value for olfactory score was set at 1.5 (dotted line), because very few individuals of the 3 fish types have scores above this value in control experiments when water is perfused in the box on both sides. The percentages in black and grey indicate the proportion of responders/non-responders, respectively. The colored lines of responders are bright, those of non-responders are pale.

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Reviewer #1 (Public Review):

Summary:
The authors posed a research question about how an animal integrates sensory information to optimize its behavioral outputs and how this process evolved. Their data (behavioral output analysis with detailed categories in response to the different odors in different concentrations by comparing surface and cave populations and their hybrid) partially answer this tough question. They built a new low-disturbance system to answer the question. They also found that the personality of individual fish is a good predictor of behavioral outputs against odor response. They concluded that cavefish evolved to specialize their response to alanine and histidine while surface fish are more general responders, which was supported by their data.

Strengths:
With their new system, the authors could generate clearer results without mechanical disturbances. The authors characterize multiple measurements to score the odor response behaviors, and also brought a new personality analysis. Their conclusion that cavefish evolved as a specialist to sense alanine and histidine among 6 tested amino acids was well supported by their data.

Weaknesses:
The authors posed a big research question: How do animals evolve the processes of sensory integration to optimize their behavioral outputs? I personally feel that, to answer the questions about how sensory integration generates proper (evolved) behavior, the authors at least need to show the ecological relevance of their response. For the alanine/histidine preference in cavefish, they need data for the alanine and other amino acid concentrations in the local cave water and compare them with those of surface water.

Also, as for "personality matters", I read that personality explains a large variation in surface fish. Also, thigmotaxis or wall-following cavefish individuals are exceeded to respond well to odorants compared with circling and random swimming cavefish individuals. However, I failed to understand the authors' point about how much percentages of the odorant-response variations are explained (PVE) by personality. Association (= correlation) was good to show as the authors presented, but showing proper PVE or the effect size of personality to predict the behavioral outputs is important to conclude "personality is matter"; otherwise, the conclusion is not so supported.

From the above, I recommend the authors reconsider the title also their research questions well. At this moment, I feel that the authors' conclusions and their research questions are a little too exaggerated, with less supportive evidence.

Also, for the statistical method, Fisher's exact test is not appropriate for the compositional data (such as Figure 2B). The authors may quickly check it at https://en.wikipedia.org/wiki/Compositional_data or https://www.annualreviews.org/doi/pdf/10.1146/annurev-statistics-042720-124436.

The authors may want to use centered log transformation or other appropriate transformations (R-package could be: https://doi.org/10.1016/j.cageo.2006.11.017). According to changing the statistical tests, the authors' conclusion may not be supported.

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Reviewer #2 (Public Review):

In their submitted manuscript, Blin et al. describe differences in the olfactory-driven behaviors of river-dwelling surface forms and cave-dwelling blind forms of the Mexican tetra, Astyanax mexicanus. They provide a dataset of unprecedented detail, that compares not only the behaviors of the two morphs but also that of a significant number of F2 hybrids, therefore also demonstrating that many of the differences observed between the two populations have a clear (and probably relatively simple) genetic underpinning.

To complete the monumental task of behaviorally testing 425 six-week-old Astyanax larvae, the authors created a setup that allows for the simultaneous behavioral monitoring of multiple larvae and the infusion of different odorants without introducing physical perturbations into the system, thus biasing the responses of cavefish that are particularly fine-tuned for this sensory modality. During the optimization of their protocol, the authors also found that for cave-dwelling forms one hour of habituation was insufficient and a full 24 hours were necessary to allow them to revert to their natural behavior. It is also noteworthy that this extremely large dataset can help us see that population averages of different morphs can mask quite significant variations in individual behaviors.

Testing with different amino-acids (applied as relevant food-related odorant cues) shows that cavefish are alanine- and histidine-specialists, while surface fish elicit the strongest behavioral responses to cysteine. It is interesting that the two forms also react differently after odor detection: while cave-dwelling fish decrease their locomotory activity, surface fish increase it. These differences are probably related to different foraging strategies used by the two populations, although, as the observations were made in the dark, it would be also interesting to see if surface fish elicit the same changes in light as well.

Further work will be needed to pinpoint the exact nature of the genetic changes that underlie the differences between the two forms. Such experimental work will also reveal how natural selection acted on existing behavioral variations already present in the SF population.

It will be equally interesting, however, to understand what lies behind the large individual variation of behaviors observed both in the case surface and cave populations. Are these differences purely genetic, or perhaps environmental cues also contribute to their development? Does stochasticity provided by the developmental process has also a role in this? Answering these questions will reveal if the evolvability of Astyanax behavior was an important factor in the repeated successful colonization of underground caves.

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Reviewer #3 (Public Review):

Summary:
The paper explores chemosensory behaviour in surface and cave morphs and F2 hybrids in the Mexican cavefish Astyanax mexicanus. The authors develop a new behavioural assay for the long-term imaging of individual fish in a parallel high-throughput setup. The authors first demonstrate that the different morphs show different basal exploratory swimming patterns and that these patterns are stable for individual fish. Next, the authors test the attraction of fish to various concentrations of alanine and other amino acids. They find that the cave

morph is a lot more sensitive to chemicals and shows directional chemotaxis along a diffusion gradient of amino acids. For surface fish, although they can detect the chemicals, they do not show marked chemotaxis behaviour and have an overall lower sensitivity. These differences have been reported previously but the authors report longer-term observations on many individual fish of both morphs and their F2 hybrids. The data also indicate that the observed behavior is a quantitative genetic trait. The approach presented will allow the mapping of genes' contribution to these traits. The work will be of general interest to behavioural neuroscientists and those interested in olfactory behaviours and the individual variability in behavioural patterns.

Strengths:
A particular strength of this paper is the development of a new and improved setup for the behavioural imaging of individual fish for extended periods and under chemosensory stimulation. The authors show that cavefish need up to 24 h of habituation to display a behavioural pattern that is consistent and unlikely to be due to the stressed state of the animals. The setup also uses relatively large tanks that allow the build-up of chemical gradients that are apparently present for at least 30 min.

The paper is well written, and the presentation of the data and the analyses are clear and to a high standard.

Weaknesses:
One point that would benefit from some clarification or additional experiments is the diffusion of chemicals within the behavioural chamber. The behavioural data suggest that the chemical gradient is stable for up to 30 min, which is quite surprising. It would be great if the authors could quantify e.g. by the use of a dye the diffusion and stability of chemical gradients.

The paper starts with a statement that reflects a simplified input-output (sensory-motor) view of the organisation of nervous systems. "Their brains perceive the external world via their sensory systems, compute information and generate appropriate behavioral outputs." The authors' data also clearly show that this is a biased perspective. There is a lot of spontaneous organised activity even in fish that are not exposed to sensory stimulation. This sentence should be reworded, e.g. "The nervous system generates autonomous activity that is modified by sensory systems to adapt the behavioural pattern to the external world." or something along these lines.

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